Simulation and Performance Evaluation – Homework Report

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https://github.com/casagrandealessandro/DES-MM1-Queue-SPE

1 Introduction

The goal of this report is to analyze the average delay in two different queuing systems using simulation. The report will be divided into 2 major parts, following the structure of the homework:

1. in the first exercise, the accuracy of the simulator with respect to the theoretical average value for a generic M/M/1 queue will be assessed;
2. in the second exercise, the performance of the same queue system under a different service time distribution, drawn through rejection sampling, will be analyzed.

2 Simulator and general assumptions

The Discrete Event Simulator for the M/M/1 queue has been implemented in Python, using a priority queue (min-heap) to manage the events arrivals and a FIFO queue to manage the queue of the single server.
In order to ensure for the long run solution analysis, the simulations will be run for a sufficient amount of time, in order to have the majority of the simulation to be in the steady state, while neglecting the transient. Instead of implementing only one very long run, it has been decided to employ the *Independent Replications* method, consisting of many (in this case 30), sufficiently long runs, each of them with a different random seed. In this way, these assumptions for the correct computation of confidence intervals formulas are fulfilled:

- the average delays from each run are independent (due to the different seeds) and identically distributed (due to the same distributions and initial conditions for each run);
- the average delays from each run are approximately normally distributed in accordance with the *Central Limit Theorem*, valid for a sufficiently high number of runs. This can be verified through *Q-Q Plots*: a representative example is shown in Figure 1, where the quantiles align more or less on the bisector for all simulated scenarios.

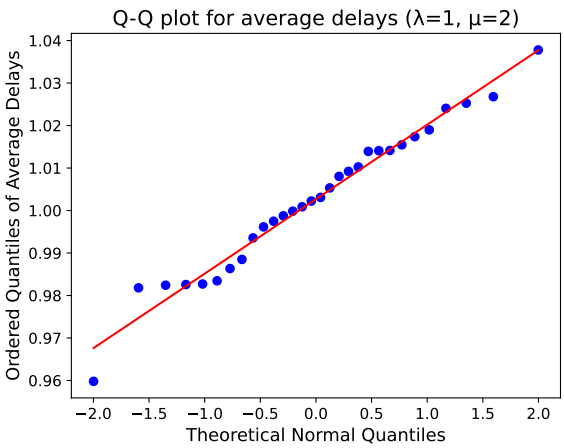


Figure 1: Q-Q Plot to verify normality

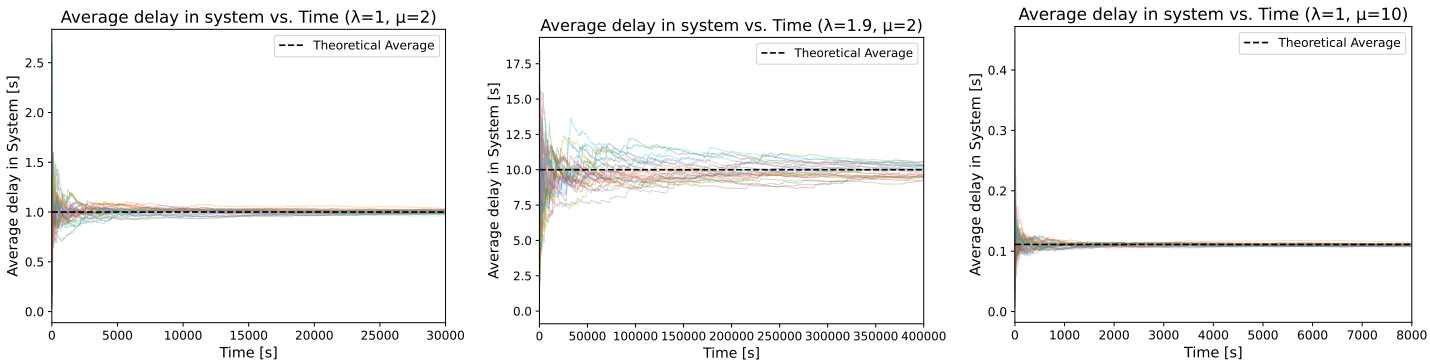
3 Exercise 1

In a generic M/M/1 queue, arrivals and departures follow a Poisson process, with parameters λ and μ respectively. Once the parameters have been set, the queue simulation has been run multiple times, and the average time spent

by a packet in the system, together with a confidence interval, have been computed. Three different scenarios have been evaluated, in particular:

1. default scenario: $\lambda = 1$, $\mu = 2$;
2. high load scenario: $\lambda = 1.9$, $\mu = 2$;
3. low load scenario: $\lambda = 1$, $\mu = 10$.

The results of the simulation are shown in Figure 2, where all the 30 runs are plotted at the same time. Please note that the axis scales are different for each plot, both in the abscissas and ordinates.



(a) Default scenario (b) High load scenario (c) Low load scenario

Figure 2: Average delay in system for the different scenarios (exercise 1)

As can be seen, there is an initial transient, where the system is still affected by its initial conditions (and thus the metrics are biased), but then as the time tends to the maximum simulated time, which has been configured depending on the parameters of the simulation, i.e. on arrivals and departures distribution, the system reaches the steady state and the metric stabilizes around the expected theoretical average $1/(\mu - \lambda)$.
From the plots, it is possible to compare the rate at which the system stabilizes, i.e. the length of the transient, as well as the different values in the average time spent in the system. In particular:

- the transient duration is much longer in the high load scenario, while it's almost negligible in the low load scenario, and this justifies the choice on the duration of the whole simulation, accordingly to the assumptions explained before;
- as the load of the system increases, it's more probable to see longer server queue, and thus the average time spent in the system is obviously longer, in accordance with the theoretical value.

These results are also visible in the statistics, reported in Table 1.

Scenario	λ	μ	**Theoretical value [s]**	**Simulated value [s]**	**C.I. (95%) [s]**
Default	1	2	1.000	1.003	[0.996, 1.009]
High Load	1.9	2	10.000	9.884	[9.727, 10.042]
Low Load	1	10	0.111	0.111	[0.110, 0.111]

Table 1: Statistics for the different scenarios (exercise 1)

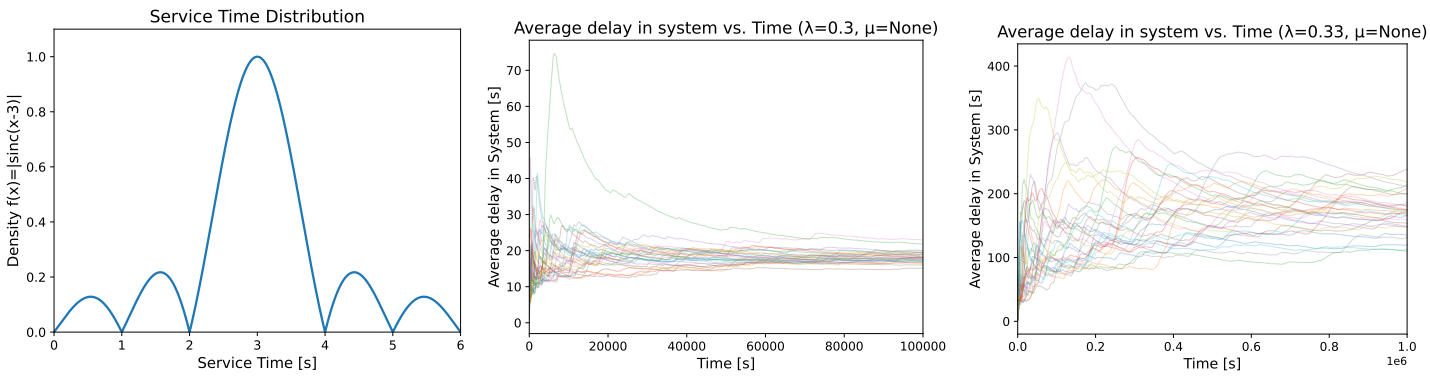
As expected from the Confidence Intervals theory, all the theoretical steady state values fall within the interval, as they should do with 95% probability.
Moreover, when the system is highly loaded, the Confidence Interval width is much wider due to the higher variance, while in the low load case the system is very stable and thus the Confidence Interval is very precise.

4 Exercise 2

The service time distribution is now

$$f(x) = A \left| \frac{\sin(\pi(x-3))}{\pi(x-3)} \right| = A \left| \text{sinc}\,(x-3) \right|, \qquad 0 \le x \le 6,$$

which has the shape reported in Figure 3a. In order to draw random values from that distribution, rejection sampling has been employed. Since the function is centered in $x = 3$ with a maximum value of $f(3) = A = 1$, a bounding box with coordinates $(0;0), (6;0), (6;1), (0;1)$ can be used. In this way, the rejection sampling simply resorts to picking a random number $U_1 \sim U[0,6]$, another random number $U_2 \sim U[0,1]$, and accepting U_1 as the sample only if $U_2 < f(U_1)$. The results would have been the same even in the case $A \neq 1$, as it affects in the same way both the density function and the bounding box, and thus the acceptance/rejection probability will be the same in any case. Figures 3b and 3c show the result of two simulations in this scenario, setting $\lambda = 0.3$ and $\lambda = 0.33$ respectively, while Table 2 reports the statistics.



(a) Service time distribution (b) Stable scenario (c) Extreme scenario

Figure 3: Service time distribution and average delay in system for the different scenarios (exercise 2)

Scenario	λ	$1/\mathbb{E}(f)$	**Simulated value [s]**	**C.I. (95%) [s]**
Stable	0.3	≈ 0.333	18.235	[17.629, 18.841]
Extreme	0.33	≈ 0.333	170.723	[159.012, 182.433]

Table 2: Statistics for the different scenarios (exercise 2)

Since $\mathbb{E}(f) = 3$ by symmetry, the average departure rate is $1/\mathbb{E}(f) \simeq 0.333 = \mu$, and therefore, to keep the system stable, the maximum arrival rate should be $\lambda < \mu$. Again, giving λ a value too close to μ puts the system in a condition in which the variance is high and reaching a stable state takes a very long time (note the time axis values, in the extreme case the convergence rate is not enough to run a simulation able to terminate in a reasonable time). In addition, outliers of the exponential arrival times distribution may have a dramatic effect that cannot be compensated by the service times distribution.

5 Conclusions

In conclusion, the simulation has validated the theoretical models, highlighting how the factors λ and μ affect both the convergence rate of the system and the average time spent in the system, as well as the impact of changing the service times distribution on the stability of the system.